

FW Subsystem Test Plan

Test parameters

- Channels under test: I and Q only
- Sampling rate: 100 Hz
- FFT size per channel: 512 samples
- FFT bin spacing: $\Delta f = 100 / 512 \approx 0.195$ Hz
- Displayed spectrum range: -4 Hz to +4 Hz
- Numerical peak frequency readout is required
- DC offset for all injected analog signals: 1.5 V
- Dynamic tests shall be executed at two amplitudes:
 - A = 50 mVpp
 - A = 3.0 Vpp
- To avoid unnecessary spectral leakage during subsystem verification, use exact FFT-bin frequencies:
 - $2 \cdot \Delta f \approx 0.391$ Hz
 - $6 \cdot \Delta f \approx 1.172$ Hz
 - $10 \cdot \Delta f \approx 1.953$ Hz
 - $16 \cdot \Delta f \approx 3.125$ Hz
 - $19 \cdot \Delta f \approx 3.711$ Hz

Common acceptance criteria

- Bin-centered tests pass if the detected dominant peak lies within **± 1 bin = ± 0.195 Hz** of the expected frequency
- Off-bin tests pass if the detected dominant peak lies at the nearest bin or otherwise within **± 1 bin** of the true frequency
- Signed tests pass only if the dominant peak remains on the correct side of 0 Hz
- Single-channel tests pass if the expected symmetric peaks at **$+f$** and **$-f$** are visible within **± 1 bin**
- Stability means that the displayed spectrum and numerical peak readout remain consistent during a **30 s observation time**
- Minor adjacent-bin leakage is acceptable if the expected peak remains dominant and no wrong-side dominance occurs

Test Plan

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
FW-01	Baseline DC / zero-frequency sanity	Apply I = 1.5 V DC, Q = 1.5 V DC	Spectrum shows only DC / near-0 content. No stable discrete peaks away from 0 Hz in the displayed band. No clipping, no stable spurious mirrored peaks, and stable behavior	as expected	passed	---
FW-02	Signed frequency, positive, low band	Apply $I(t) = 1.5 \text{ V} + (A/2) * \cos(2\pi f t)$, $Q(t) = 1.5 \text{ V} + (A/2) * \sin(2\pi f t)$ with f = 0.391 Hz . Run once at A = 50 mVpp and once at A = 3.0 Vpp	Dominant spectral peak is at +0.391 Hz ± 1 bin . The negative-frequency side does not become dominant. Spectrum and numerical peak readout are stable	+0.391 Hz	passed	---
FW-03	Signed frequency, positive, mid band	Same as FW-02, with f = 1.172 Hz . Run at A = 50 mVpp and A = 3.0 Vpp	Dominant peak at +1.172 Hz ± 1 bin . No sign inversion. Spectrum and numerical peak readout are stable	+1.172 Hz	passed	---
FW-04	Signed frequency, positive, upper band	Same as FW-02, with f = 3.125 Hz . Run at A = 50 mVpp and A = 3.0 Vpp	Dominant peak at +3.125 Hz ± 1 bin . No sign inversion. Spectrum and numerical peak readout are stable	+3.125 Hz	passed	---
FW-05	Signed frequency, positive, near display edge	Same as FW-02, with f = 3.711 Hz . Run at A = 50 mVpp and A = 3.0 Vpp	Dominant peak at +3.711 Hz ± 1 bin and still correctly displayed inside the visible range.	+3.711 Hz	passed	---

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
			No false wraparound or wrong-side dominance. Spectrum and numerical peak readout are stable			
FW-06	Signed frequency, positive, off-bin	Same as FW-02, with f = 1.300 Hz . Run at A = 50 mVpp and A = 3.0 Vpp	Dominant peak around +1.300 Hz , at the nearest bin or otherwise within ±1 bin of the true frequency. Some leakage over adjacent bins is expected, but the negative-frequency side shall not become dominant. Spectrum and numerical peak readout are stable	+1.367 Hz	passed	---
FW-07	Signed frequency, positive, I/Q amplitude mismatch	Apply $I(t) = 1.5 \text{ V} + (A_I/2) * \cos(2\pi f t)$, $Q(t) = 1.5 \text{ V} + (A_Q/2) * \sin(2\pi f t)$ with f = 1.172 Hz . Run once with A_I = 50 mVpp , A_Q = 35 mVpp and once with A_I = 3.0 Vpp , A_Q = 2.1 Vpp	Dominant peak at +1.172 Hz ± 1 bin . Some degradation of spectral symmetry or additional minor leakage is acceptable due to the amplitude mismatch, but the negative-frequency side shall not become dominant . Spectrum and numerical peak readout are stable	+1.172 Hz	passed	---
FW-08	Signed frequency,	Apply $I(t) = 1.5 \text{ V} + (A/2) * \cos(2\pi f t)$,	Dominant spectral peak is at -0.391 Hz ± 1 bin . The	-0.391 Hz	passed	---

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
	negative, low band	$Q(t) = 1.5 \text{ V} - (A/2) * \sin(2\pi f t)$ with $f = 0.391 \text{ Hz}$. Run at $A = 50 \text{ mVpp}$ and $A = 3.0 \text{ Vpp}$	positive side does not become dominant. Spectrum and numerical peak readout are stable			
FW-09	Signed frequency, negative, mid band	Same as FW-08, with $f = 1.172 \text{ Hz}$. Run at $A = 50 \text{ mVpp}$ and $A = 3.0 \text{ Vpp}$	Dominant peak at -1.172 Hz ± 1 bin . No sign inversion. Spectrum and numerical peak readout are stable	-1.172 Hz	passed	---
FW-10	Signed frequency, negative, upper band	Same as FW-08, with $f = 3.125 \text{ Hz}$. Run at $A = 50 \text{ mVpp}$ and $A = 3.0 \text{ Vpp}$	Dominant peak at -3.125 Hz ± 1 bin . No sign inversion. Spectrum and numerical peak readout are stable	-3.125 Hz	passed	---
FW-11	Signed frequency, negative, off-bin	Same as FW-08, with $f = 2.700 \text{ Hz}$. Run at $A = 50 \text{ mVpp}$ and $A = 3.0 \text{ Vpp}$	Dominant peak around -2.700 Hz , at the nearest bin or otherwise within ± 1 bin of the true frequency. Some leakage over adjacent bins is expected, but the positive-frequency side shall not become dominant. Spectrum and numerical peak readout are stable	-2.734 Hz	passed	---
FW-12	Signed frequency, negative, I/Q amplitude mismatch	Apply $I(t) = 1.5 \text{ V} + (A_I/2) * \cos(2\pi f t)$, $Q(t) = 1.5 \text{ V} - (A_Q/2) * \sin(2\pi f t)$ with $f = 1.172 \text{ Hz}$. Run once with $A_I = 50 \text{ mVpp}$, $A_Q = 35 \text{ mVpp}$ and	Dominant peak at -1.172 Hz ± 1 bin . Some degradation of spectral symmetry or additional minor leakage is acceptable due to the amplitude	-1.172 Hz	passed	---

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
		once with A_I = 3.0 Vpp , A_Q = 2.1 Vpp	mismatch, but the positive-frequency side shall not become dominant . Spectrum and numerical peak readout are stable			
FW-13	Single-channel injection on I only (unsigned behavior)	Apply $I(t) = 1.5 \text{ V} + (A/2) * \cos(2*\pi*f*t)$ with f = 1.953 Hz . Keep Q = 1.5 V DC . Run at A = 50 mVpp and A = 3.0 Vpp	Spectrum shows the expected symmetric pair at +1.953 Hz and -1.953 Hz , both within ±1 bin . No false signed dominance shall be interpreted from this test. Spectrum and numerical peak readout are stable	+1.953 Hz / -1.953 Hz	passed	---
FW-14	Single-channel injection on Q only (unsigned behavior)	Keep I = 1.5 V DC . Apply $Q(t) = 1.5 \text{ V} + (A/2) * \cos(2*\pi*f*t)$ with f = 1.953 Hz . Run at A = 50 mVpp and A = 3.0 Vpp	Spectrum shows the expected symmetric pair at +1.953 Hz and -1.953 Hz , both within ±1 bin . No false signed dominance shall be interpreted from this test. Spectrum and numerical peak readout are stable	+1.953 Hz / -1.953 Hz	passed	---